UTKARSH SHARMA

Software Systems & Data Science · 2nd-Year B.Tech Undergraduate

GitHub: github.com/shutkarshX • LinkedIn: in/utkarsh-sharma • Live Production: tein.onrender.com • Greater Noida, India

EDUCATION

Noida Institute of Engineering & Technology (NIET) Aug 2025 — Apr 2029

Bachelor of Technology (B.Tech) in Computer Science & Engineering (Specialization: Data Science) Greater Noida, India

Core Coursework: Data Structures & Algorithms, Operating Systems, Systems Programming, Database Management Systems, Object-Oriented Design (Java), Computer Networks, Statistical Data Analysis.

EXPERIENCE

Prodigy InfoTech Sep 2026 — Oct 2026

Data Science Intern · Remote Internship

- Completed a 5-task data science internship pipeline covering predictive modeling, exploratory analysis, and NLP classification.
- Trained and tuned **Decision Tree classifiers** on customer demographic datasets to predict purchase/subscription propensity, optimizing precision and recall across feature thresholds.
- Conducted in-depth **Exploratory Data Analysis (EDA)** on demographic and Titanic survival records using Pandas and Seaborn to discover key correlations and survival patterns.
- Implemented text preprocessing and sentiment mining pipelines in Python to categorize social media feedback trends.

KEY TECHNICAL PROJECTS

TEIN — Attendance Intelligence & Campus Hub Platform [tein.onrender.com] [repo] Python, Flask, Docker, PWA, Render

- Engineered and deployed a mobile-first university platform on Render to automate attendance tracking and student society discovery.
- Implemented **Bunk & Recovery algorithms** calculating safe class skip thresholds and mandatory attendance recovery plans to safeguard the 75% institutional compliance requirement.
- Built a forward-looking **What-If Simulator** enabling students to model hypothetical absence impacts across future semester dates.
- Configured offline-capable **PWA architecture** (`sw.js`) and an integrated directory indexing 20+ university societies and technical clubs.

RESQ-AI — Emergency Operations & Disaster Command Center [repo] FastAPI, React, Gemini AI, Leaflet GIS

- Architected an emergency coordination dashboard integrating **Google Gemini AI** to triage citizen distress calls and classify urgency severity.
- Rendered real-time geospatial maps via **Leaflet GIS** plotting active incident pins, rescue convoy routes, and relief shelter capacity.
- Implemented high-concurrency async endpoints in FastAPI with Pydantic type validation and automated OpenAPI/Swagger documentation.

Layer-Flow — Transparent Screen Annotation & Desktop Canvas [repo] Rust, Tauri 2, React, Win32 API

- Developed a native Windows transparent desktop overlay in Rust and Tauri 2 with sub-50MB memory footprint and zero UI latency.
- Integrated low-level **Win32 API hooks** for global system hotkeys, click-through mode toggling, and multi-monitor screen drawing.

CPU Scheduling Algorithms Lab & Systems Simulator [repo] Python, Operating Systems, Pytest

- Implemented core OS CPU scheduling algorithms: FCFS, SJF (Preemptive & Non-Preemptive), Priority Scheduling, and Round Robin.
- Authored an exhaustive test suite with **166 automated unit tests** verifying turnaround time, wait time, and starvation resilience.

TECHNICAL SKILLS

Languages:	Python, Rust, JavaScript, TypeScript, C (C99), Java, SQL, R, HTML/CSS
Frameworks & Libraries:	FastAPI, Flask, React, Vite, Tauri 2, Pandas, Scikit-Learn, Seaborn, NumPy, Leaflet
Systems & Cloud:	Docker, Linux / Bash, Progressive Web Apps (PWA), Win32 API, Render Cloud, Git / GitHub Actions
Testing & Methodologies:	Pytest (190+ automated tests across repos), REST APIs, Asyncio, MVC Architecture, Agile